Permutations & Combinations: Formula Sheet

1. Basics

Product / sum: $m \times n$ / $m + n$

Union: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Three sets:

$$\sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$$

Permutation:

$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

Combination:

$$^{n}C_{r} = \frac{n!}{r!(n-r)!}$$

Relation: $^{n}P_{r} = {}^{n}C_{r}\, r!$

Symmetry: $^{n}C_{r} = {}^{n}C_{n-r}$

Equal: $^{n}C_{x} = {}^{n}C_{y} \Rightarrow x = y$ or $x + y = n$

Pascal: $^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$

Ratio:

$$\frac{^{n}C_{r}}{^{n}C_{r-1}} = \frac{n-r+1}{r}$$

Maximum $^{n}C_{r}$: $r = \frac{n}{2}$ or $\frac{n\pm1}{2}$

2. Factorial

Recurrence: $n! = n\,(n-1)!$

Double:

$$(2n)! = 2^{n}\, n!\, [1 \cdot 3 \cdots (2n-1)]$$

Power of p:

$$E_{p}(n!) = \sum \left\lfloor \frac{n}{p^{k}} \right\rfloor$$

Zeroes at the end: $E_{5}(n!)$

Telescoping: $r \cdot r! = (r+1)! - r!$

3. Numbers and words

Repetition: n^{r}; no repetition: $^{n}P_{r}$

Increasing digits: $^{9}C_{n}$; decreasing: $^{10}C_{n}$

Divisible by 4: last two digits

Divisible by 11: odd − even place sums

Alike objects:

$$\frac{n!}{p!\, q!\, r!}$$

Together (string): $(n-k+1)!\, k!$

Apart (gap): $(n-r)!\, {}^{n-r+1}P_{r}$

Alternate: $2\, n!\, n!$ or $(n+1)!\, n!$

Fixed relative order of k:

$$\frac{n!}{k!}$$

Adjacent different: $k(k-1)^{n-1}$

Circle, adjacent different: $(k-1)^{n} + (-1)^{n}(k-1)$

4. Geometry

Lines: $^{n}C_{2} - {}^{p}C_{2} + 1$

Triangles: $^{n}C_{3} - {}^{p}C_{3}$

Quadrilaterals:

$$^{n}C_{4} - {}^{p}C_{4} - {}^{p}C_{3}\, {}^{n-p}C_{1}$$

Diagonals:

$$\frac{n(n-3)}{2}$$

Chord intersections inside: $^{n}C_{4}$

Intersections: $-{}^{p}C_{2}$ parallel, $-{}^{q}C_{2} + 1$ concurrent

Rectangles: $^{m+1}C_{2}\, {}^{n+1}C_{2}$

Squares: $\sum (m-k+1)(n-k+1)$

AP triples: $\frac{(n-1)^{2}}{4}$ (odd n), $\frac{n(n-2)}{4}$ (even n)

Polygon, no common side:

$$\frac{n}{n-r}\, {}^{n-r}C_{r}$$

Station (row): $^{n-r+1}C_{r}$

5. Groups and circles

Groups p, q, r:

$$\frac{n!}{p!\, q!\, r!}$$

Equal unlabelled:

$$\frac{(kp)!}{(p!)^{k}\, k!}$$

Equal labelled:

$$\frac{(kp)!}{(p!)^{k}}$$

Each gets one: $2^{n} - 2$, $3^{n} - 3 \cdot 2^{n} + 3$

Onto: $\sum (-1)^{rk}\, {}^{k}C_{r}(k-r)^{n}$

Circular: $(n-1)!$; r of n:

$$\frac{^{n}P_{r}}{r}$$

Garland / necklace:

$$\frac{(n-1)!}{2}$$

Circle, together: $(n-k)!\, k!$

Circle, apart: $(m-1)!\, {}^{m}P_{r}$

6. Selections and divisors

All selections: $(p+1)(q+1)(r+1)2^{k} - 1$

Divisors: $d(N) = (a+1)(b+1)(c+1)$

Odd divisors: $(b+1)(c+1)$

Even divisors: $a(b+1)(c+1)$

Sum:

$$\sigma(N) = \prod \frac{p^{a+1}-1}{p-1}$$

Multiples of M: sum $= M\,\sigma(N/M)$

Two factors: $\frac{d}{2}$ or $\frac{d+1}{2}$

Coprime two factors: 2^{k-1}

Product of divisors: $N^{d/2}$

Pairs with LCM N: $\prod(2a_i+1)$

Euler:

$$\varphi(N) = N \prod \left(1 - \frac{1}{p}\right)$$

7. Paths, derangements, sums

Grid: ${}^{a+b}C_a$

With r diagonals:

$$\frac{(a+b-r)!}{r!(a-r)!(b-r)!}$$

$$D_n = n! \sum \frac{(-1)^r}{r!}$$

$$D_n = (n-1)(D_{n-1} + D_{n-2})$$

$$D_1..D_6 = 0, 1, 2, 9, 44, 265$$

Exactly r fixed: ${}^nC_r\, D_{n-r}$

Digit sum: $f \cdot \Sigma d \cdot 11\cdots1$

8. Alike objects and solutions

Beggars: ${}^{n+p-1}C_{p-1}$

Each at least one: ${}^{n-1}C_{p-1}$

Minimum m_i: ${}^{n-\sum m_i+p-1}C_{p-1}$

$x_1 + \cdots + x_k \le n$: ${}^{n+k}C_k$

$x_1 x_2 \cdots x_k = N$: $\prod {}^{a+k-1}C_{k-1}$

Bounded factor: $x^a + \cdots + x^b$

Alike p: $1 + x + \cdots + x^p$

9. Extra reading

Fibonacci: $F_n = F_{n-1} + F_{n-2}$

Partitions: $P(n,k) = P(n,k-1) + P(n-k,k)$

$P(n,2) = \lfloor \frac{n}{2} \rfloor + 1$, $P(n,3) \approx \frac{(n+3)^2}{12}$

Pentagonal:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + \cdots$$

Stirling: $S(n,k) = kS(n-1,k) + S(n-1,k-1)$

$$S(n,2) = 2^{n-1} - 1$$

Bell: $B_n = \sum S(n,k)$

$$B_0..B_6 = 1, 1, 2, 5, 15, 52, 203$$

$x_1 + 2x_2 + \cdots + kx_k = n$: $P(n,k)$ solutions